

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

DSA

COURSE 11 OF 12
CURRICULUM GUIDE

DSA for Teens

Data structures, algorithms & problem solving

GRADE BAND

Grade 9–12

DIFFICULTY

Intermediate–Advanced

DURATION

16 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

The problem-solving foundation used in technical interviews and competitive programming — taught through guided, incremental practice rather than memorized answers.

GRADE BAND	DIFFICULTY	DURATION
Grade 9–12	Intermediate–Advanced	16 weeks · 2 sessions/week

Prerequisites: Python for Kids & Teens, or Java Development for Teens.

LEARNING OUTCOMES

What Students Will Learn

- Analyze algorithm efficiency using Big-O notation
- Implement and apply core data structures: arrays, stacks, queues, linked lists, trees, hash maps
- Apply searching and sorting algorithms and understand their tradeoffs
- Use recursion to solve problems
- Apply common problem-solving patterns (two pointers, sliding window, and more)
- Solve original coding problems under time constraints

WHAT WE'LL USE

Tools & Technologies

Python or Java Online judge / practice platform Whiteboarding

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · Big-O & Algorithmic Thinking	Measuring and comparing efficiency.
02 · Arrays & Strings	Core patterns and the two-pointer technique.
03 · Searching & Sorting	Linear/binary search and common sort algorithms.

MODULE	FOCUS
04 · Recursion	Thinking recursively; base cases.
05 · Linked Lists, Stacks & Queues	Implementation and real use cases.
06 · Trees & Hash Maps	Traversals, lookups and practical applications.
07 · Problem-Solving Patterns	Sliding window, greedy strategies, an intro to dynamic programming.
08 · Capstone Problem Set & Mock Interview	Timed practice with guided walkthroughs.

LEARNING BY BUILDING

Projects Students Will Build

- Personal Algorithm Notebook — Documented solutions with clear explanations.
- Data Structures From Scratch — Custom implementations, not just library calls.
- Capstone Problem Set — 30–40 solved problems plus a mock interview session.

BEYOND THE SYNTAX

Skills Developed

Algorithmic thinking Technical problem solving Complexity analysis Interview readiness

PORTFOLIO OUTCOME

A GitHub repository of solved, well-explained problems that shows visible growth over time.

RECOMMENDED NEXT STEP

System Design for Teens